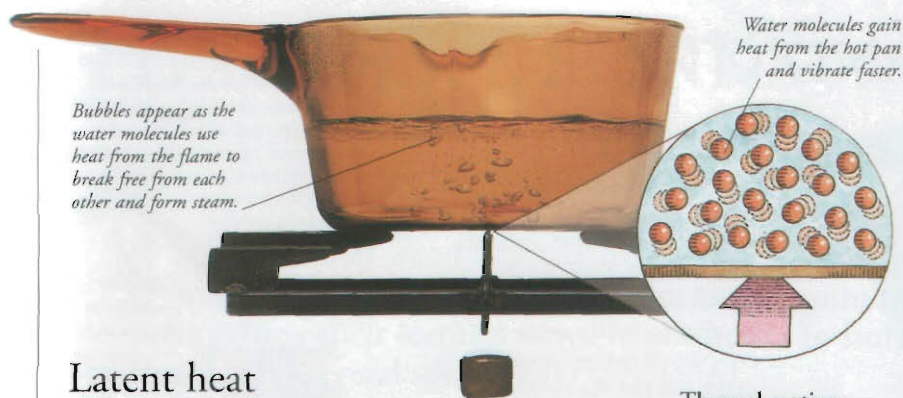




Latent heat

When a liquid is at its boiling point, an input of heat energy will not raise the liquid's temperature any further. Instead, the extra energy enables particles in the liquid to break free from each other and form a gas. This energy is called latent heat. The energy is released again if the gas condenses into a liquid. Latent heat is also absorbed when a solid melts, and released when a liquid freezes.

Thermal motion

All matter is made up of moving particles. This movement is called thermal motion. The temperature of an object is a measure of its thermal motion. Heating the object makes its particles vibrate faster and raises its temperature.

Radiation

All objects give out energy in the form of infrared rays, which are similar to X-rays. A hot object, such as a light bulb, gives out a lot of infrared rays. These rays will heat up any object that absorbs them. Dull surfaces absorb infrared rays well, but shiny surfaces reflect them. Infrared rays are invisible, but you can feel their effect. The closer you put your hand to a light bulb, the warmer it feels, because the radiation is more intense.

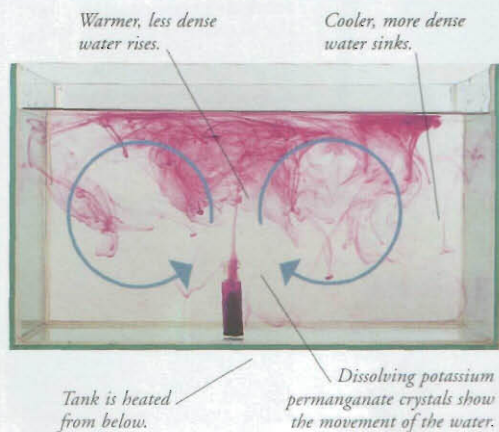


Propagator

Seeds sprout and grow more rapidly in warm conditions, so they are often planted in a tiny greenhouse called a propagator. Sunlight passes through the propagator's plastic cover and warms the seeds and soil, which radiate the heat back out again as infrared rays. The rays cannot pass through plastic, so the heat is trapped inside and the temperature rises.

Convection

The way heat travels as moving currents through a gas or a liquid is called convection. If a tank of water is heated from below, the warm water at the bottom will rise as it expands and becomes less dense. The cooler, denser water above sinks to take its place. Soon, this cooler water also warms and starts to rise, creating a circulation of water called a convection current.



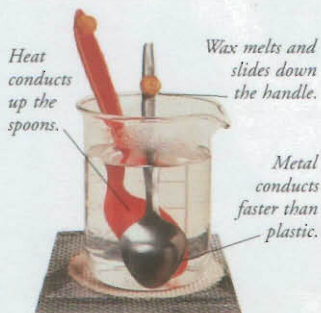
Heat conduction

When a substance is heated, its vibrating particles knock against neighbouring particles and pass on some of their thermal motion, spreading heat throughout the substance. This is called conduction. Heat always conducts from a warm substance to a cooler one. The warm air in a room can lose heat through a window. Heat conducts from the warm air to the window, and then to the cooler air outside.



Heat conductors

Some materials – called conductors – conduct heat better than others. If you press wax on to the ends of metal and plastic spoons in hot water, the wax on the metal spoon melts first. This is because metal is a better conductor.



Hot or cold?

A marble tile feels colder than a carpet at room temperature. Marble is a better conductor than carpet. Marble takes heat rapidly away from the body, making the marble tile feel much colder to the touch than the carpet.



Heat insulators

Poor conductors of heat, such as plastics, wood, cork, and air, are known as insulators. Using such materials to reduce heat loss from an object is called insulation. When these two jars are filled with water at 80°C (176°F) and left for 15 minutes, the jar covered in bubble wrap retains the most heat. Bubble wrap (plastic filled with pockets of air) is a good



Vacuum flask

A vacuum flask keeps drinks hot or cold by stopping the transfer of heat to or from the liquid. Conduction can only occur through matter, so the flask has two walls with a vacuum between them to prevent conduction. Shiny walls reflect heat radiation, while the flask is sealed with an airtight stopper made of a good insulator.

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